Part A – Theoretical Foundation

1. What is Inferential Statistics?

- **Definition:** Inferential statistics involves using data collected from a representative sample to make valid conclusions, predictions, or estimations about the broader population from which the sample was drawn. It allows researchers to generalize findings without the need to evaluate every individual in a population.

  💡 **Application Example:** Studying a sample of 200 randomly selected patients to estimate the average blood pressure of all residents across an entire city.

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2. What is Hypothesis Testing and its Components?

- **Definition:** Hypothesis testing is a formal statistical method used to evaluate the validity of a specific claim or assumption regarding a population parameter based on sample evidence.

Core Components:

- **Null Hypothesis H0:** The default assumption asserting that there is no statistical significance, no effect, or no difference between the variables being tested.
- **Alternative Hypothesis (H1):** The statement that directly contradicts the null hypothesis, asserting that there is a significant effect, presence of a relationship, or distinct difference.

  💡 **Application Example:**
  - **H0:** Smoking habits do not affect the prevalence of diabetes.
  - **H1:** Smoking habits significantly affect the prevalence of diabetes.

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3. Explain Confidence Interval and Critical Value

Confidence Interval (CI)

- **Definition:** A confidence interval provides an estimated range of values, calculated from sample data, that is likely to contain the true, unknown population parameter. This range is accompanied by a specific confidence level (typically **95%**), which represents the long-term success rate of the estimation procedure.

  💡 **Application Example:** If the average sample BMI is **24** with a **95%** confidence interval of [23, 25], it implies that the true population average BMI is highly likely to fall between **23** and **25**.

Critical Value

- **Definition:** The critical value serves as the mathematical threshold or cut-off point on a statistical distribution scale. It partitions the scale into regions of acceptance and rejection, determining whether the calculated test statistic warrants rejecting the null hypothesis.

  💡 **Application Example:** In a standardized test, if the computed sample test statistic exceeds the predetermined critical value threshold, the null hypothesis H0 is formally rejected.

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4. Define P-Value

- **Definition:** The p-value (probability value) quantifies the probability of obtaining test results at least as extreme as the observed sample results, under the foundational assumption that the null hypothesis H0 is perfectly true.

  💡 **Application Example:** A calculated p-value of **0.03** (p = 0.03) indicates there is only a **3%** probability that the observed differences occurred due to random chance. Because this is less than the standard significance threshold (\alpha = 0.05), the null hypothesis is rejected.

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5. Differentiate Type I and Type II Errors

Statistical decisions based on sample data carry inherent risks of misinterpretation, which are categorized into two distinct types of errors:

Error Type	Statistical Definition	Practical Conceptualization	Clinical Example
Type I Error	Incorrectly rejecting the null hypothesis (H0) when it is true.	A "false positive" or false alarm.	Diagnosing a completely healthy individual as having diabetes.
Type II Error	Failing to reject the null hypothesis H0	A "false negative" or missed detection.	Failing to diagnose a truly diabetic individual,

	when it is false.		classifying them as healthy.

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6. Core Statistical Tests

Choosing the correct statistical approach depends entirely on the sample dimensions, variance availability, and data types:

Statistical Test	**Core Technical Criteria**	**Practical Scenario Example**
Z-Test	Applied to large, normally distributed datasets (typically sample size $n > 30$) or when the population variance is explicitly known.	Comparing the average baseline blood pressure records between a large group of **500** men and **500** women.
T-Test	Utilized for smaller sample dimensions ($n < 30$) or when the true population variance remains entirely unknown.	Comparing the average calculated BMI of **20** regular smokers against a small control group of **20** non-smokers.
Chi-Square Test	An analytical test used to compare observed frequencies against expected frequencies within categorical variables.	Evaluating whether a categorical habit like smoking is independently associated with a diabetes diagnosis.
ANOVA *(Analysis of Variance)*	Compares the mean scores across more than two distinct independent groups by partitioning total variance into between-group and within-group	Evaluating and comparing baseline blood cholesterol levels across **5** separate age-bracket groups.

	components.	

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7. What is Covariance?

- **Definition:** Covariance measures the directional relationship between two continuous variables, indicating whether they tend to increase or decrease together.
  - **Positive Covariance:** Indicates that as one variable increases, the other variable generally tends to increase concurrently.
  - **Negative Covariance:** Indicates an inverse relationship where an increase in one variable corresponds to a decrease in the other.
  - **Core Limitation:** The raw output value is entirely dependent on the specific units of measurement used, making it difficult to assess the actual strength of the relationship.

  💡 **Application Example:** If tracking physical health reveals that as an individual's age increases, their BMI also increases, the resulting covariance calculation will yield a positive value.

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8. What is Correlation?

- **Definition:** Correlation is a standardized, scaled adaptation of covariance that evaluates both the strength and the directional nature of the linear relationship between two variables, bounding the output cleanly between -1.0 and +1.0.

  💡 **Application Example:** A calculated correlation coefficient of **0.9** between age and BMI signifies a highly resilient, positive linear relationship, indicating that increases in age map predictably to increases in BMI values.